

Adjustments

Shut off power and disconnect electrical connections before making any adjustments.

Height Adjustment and Leveling

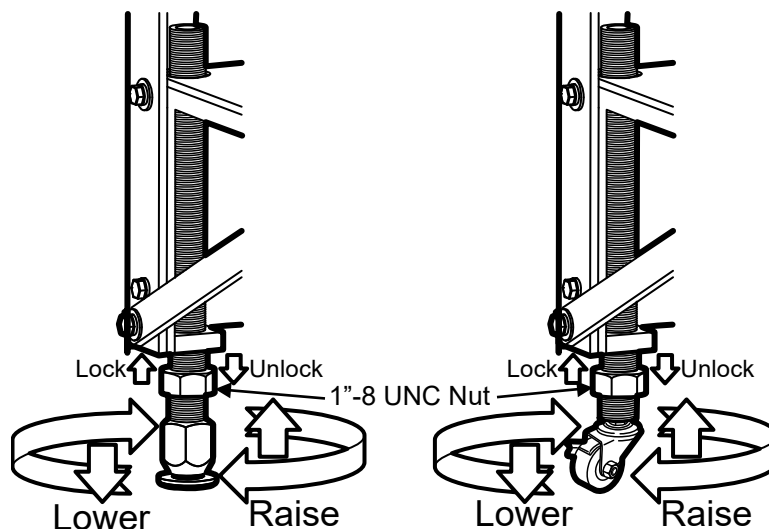
All four legs at the corners of the conveyor are adjustable for height and leveling, and two mechanisms allow for height adjustment: threaded-rod legs and leg-extension angle brackets.

Threaded-rod legs: threaded-rod legs are adjustable for their total length (less approximately $\frac{1}{2}$ inch of threads required to secure the legs in the bottom of leg-extension angle brackets) and provide for fine adjustment to level the conveyor at the required conveyor height.

Leg-extension angle brackets: Extending or retracting the leg-extension angles provide five inches (5") additional height when extended, or less five inches (5") when retracted. When extending the leg-extension angles, it is recommended to reposition the leg brace strap for stability. See the illustrations on the following page.

Raising or Lowering by Small Increments or Leveling the Conveyor

For fine-adjustment of the leg length for fine adjustment of height or leveling the conveyor, first loosen the 1"-8 UNC standard nut, if applicable, and then twist the threaded-rod leg into or out of the conveyor leg angle bracket.

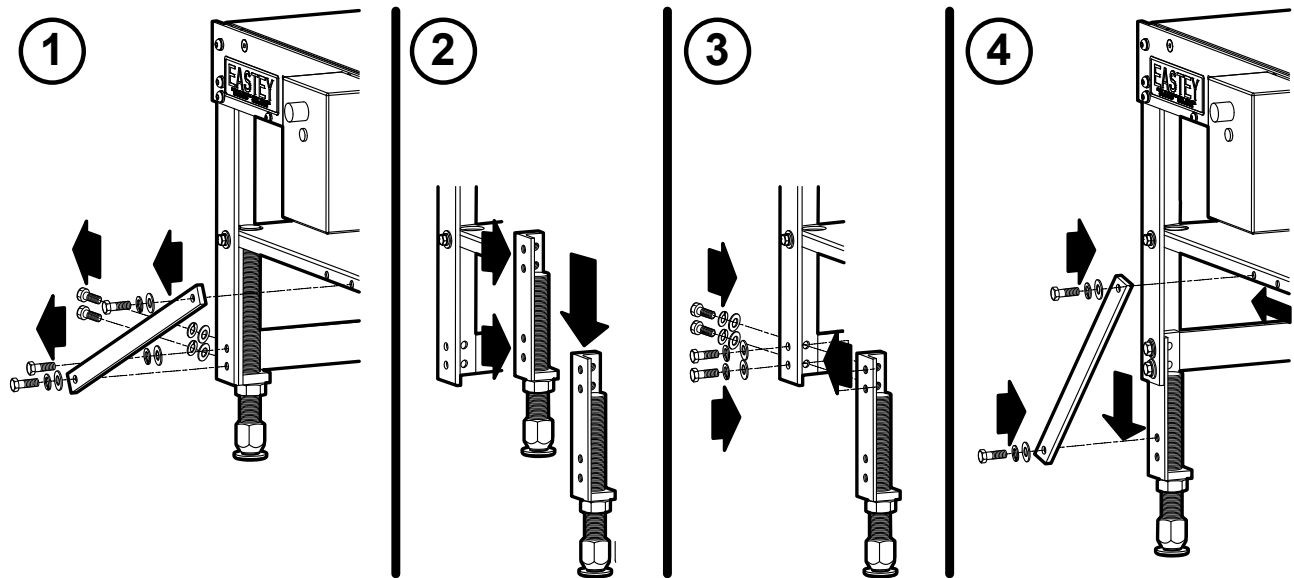


Note: An open-end wrench or channel-style wrench with a 1- $\frac{1}{2}$ " opening is required for the 1"-8 UNC standard nuts.

Note: Thread-locking compound is used to secure the feet or optional casters to the leveling leg.

Leg Extension Angle Brackets

Leg-extension angle brackets in each of the four legs of the conveyor frame provide five inches (5") additional height when extended, or less five inches (5") when retracted. When extending or retracting the leg-extension angles, it is recommended to reposition the leg brace strap accordingly for stability.

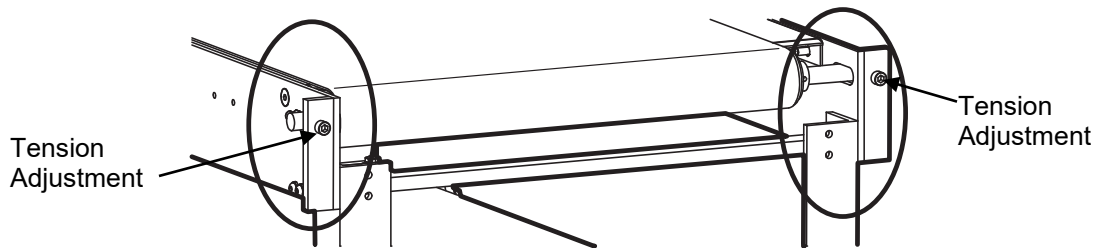


1. Temporarily remove and save the hardware that fastens the leg extension angle bracket and leg brace bar. Remove both ends of the leg brace bar, as the brace bar will be repositioned to other holes when it is refastened to the leg extension angle bracket.
2. Reposition the leg extension angle to align the upper holes in the leg extension angle with the bottom holes in the vertical leg of the conveyor base frame.
3. With the leg extension angle bracket extended, use the same hardware removed previously to refasten the leg extension angle bracket to the conveyor base frame.
4. Reposition the leg brace bar as shown, to align the top hole with the hole closer to the corner in the side brace of the conveyor base frame, and to use the upper of the two holes near the bottom of the leg extension angle bracket.

After changing the conveyor height by extending or retracting the leg extension angle brackets, return to the instructions for Raising or Lowering by Small Increments or Leveling the Conveyor

Conveyor Tension Adjustment

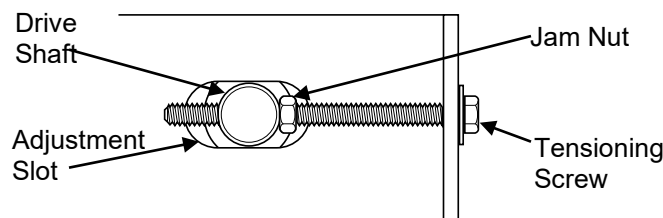
Adjustment of conveyor tension is made by tightening or loosening the conveyor tension adjustment screws. There is a tension adjustment screw on each side of the conveyor at the right and left ends of the idler shaft. Jam nuts secure the tension adjustment.



Ideally the conveyor shaft should be adjusted so that when correct tension is achieved, the shaft is near the center of the adjustment slot. Before beginning adjustment, measure these distances to position the conveyor shaft near the center of the adjustment slot when proper tension is achieved.

Note: It is necessary to run the conveyor to verify that it is tracking correctly and not pulling to either side and wearing against the side frame. Make small adjustments, one at a time on each side, until proper belt tension is achieved and belt is tracking properly.

1. Loosen the jam nuts on the adjustment screws away from the conveyor shaft.



2. Use a wrench on the screw head to adjust tension.
 - Turn the adjustment screw counterclockwise to loosen tension.
 - Turn the adjustment screw clockwise to increase tension.
3. Ensure tracking of belt is centered between frames so the belt does not wear on either side frame. The belt will pull to the side where tension is tighter.
4. When the conveyor tension is correctly adjusted, re-tighten the jam nuts against the drive shaft to lock-in tension setting.

Advanced Speed and Control Function Adjustment

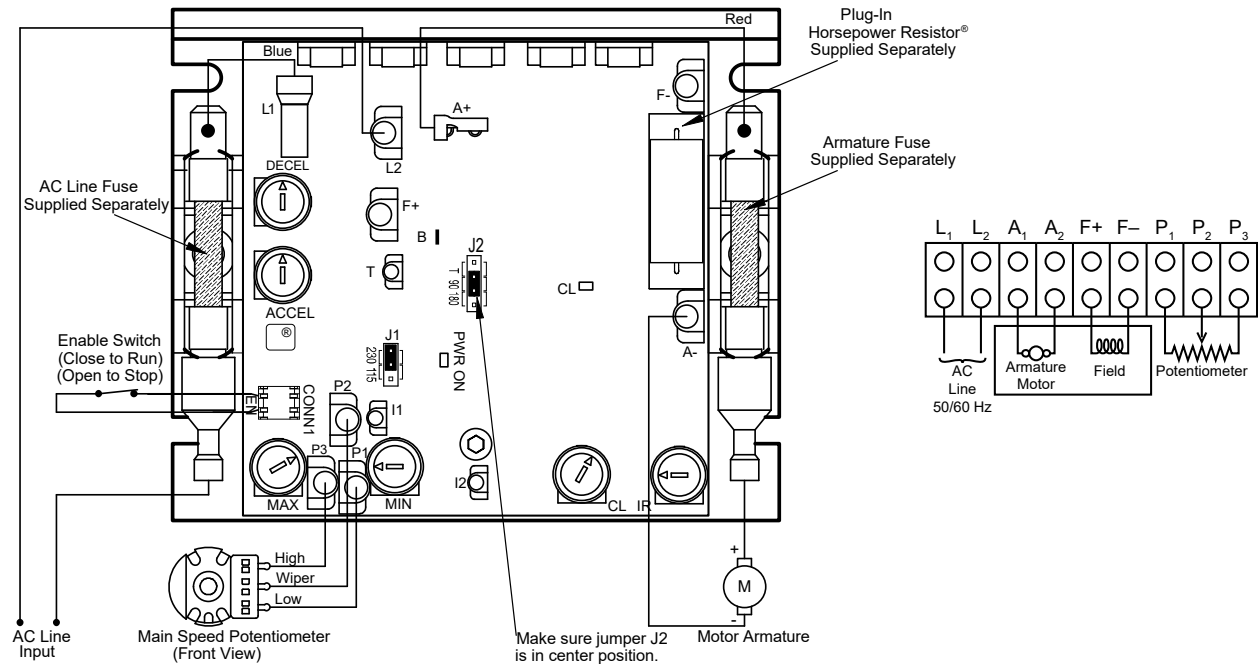
The following illustration shows the D.C. board housed in the conveyor Control Box. Adjustments explained in this section refer to adjustments made by turning potentiometers on this board.

Basic KBMM™ Controller Board Connection Diagram

KBMM™ with Barrier Terminal Kit

CONTROL LAYOUT & GENERAL CONNECTION DIAGRAM (Model KBMM-225D Shown)

(Note: Control is set for 208/230 VAC line input, 0-180 VDC output with armature feedback)

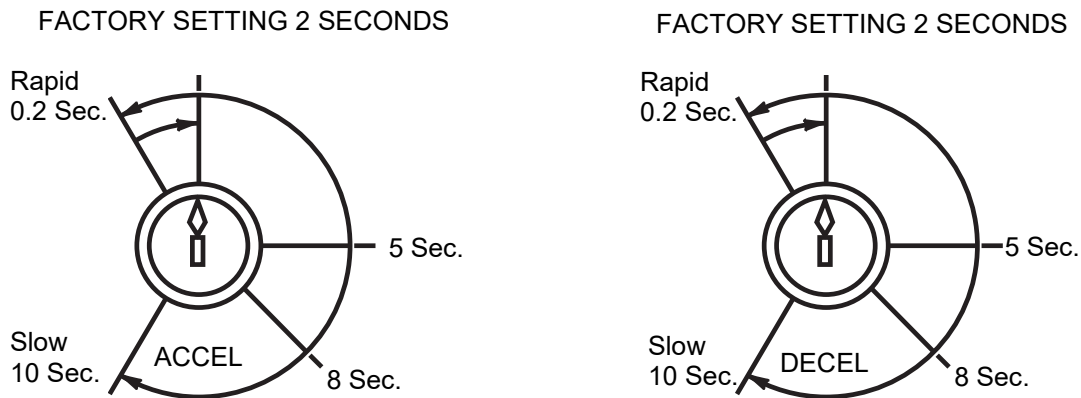


For more information refer to the KBMM™ *Installation and Operation Manual* (provided by the D.C. board manufacturer).

The Multi-Drive™ has been factory adjusted to provide zero (0) to full-speed range using the Speed Control Dial. Minimum and maximum speed control trim-potentiometers (trimpots) are provided to increase the minimum speed or decrease the maximum speed if necessary. An acceleration start trimpot is set to provide motor acceleration from zero (0) to full speed over a time period of 2 seconds (approximately) each time AC power is applied. The Current Limiting (CL, or torque output) adjustment is factory set to approximately one-and-one-half ($1\frac{1}{2}$) times the motor rating. The IR Compensation (IR) is factory adjusted to provide near-optimum motor regulation under normal operation.

Note: In order for the IR comp and CL trimpot settings to be correct, the proper Plug-In Horsepower Resistor® must be installed for the particular motor and input voltage being used. Do not attempt to change the setting of the trimpots unless absolutely necessary since they are factory adjusted to near optimum settings.

The following procedure, presented in order of adjustment sequence should be used when readjusting all trimpot functions.



- A. **Acceleration, Start, and Deceleration.** ACCEL and DECEL trimpots are located near the left side of the speed control module. If the ACCEL and/or DECEL are to be readjusted to different times, adjust trimpots according to the illustration above.
- B. **Maximum Speed Adjustment.** Turn Speed Control Dial to full speed (maximum clockwise position). Adjust MAX speed trimpot to new desired setting.

Note: Do not attempt to adjust the maximum speed above the rated motor RPM since unstable motor operation may occur. For moderate changes to the maximum speed, there will be slight effect on the minimum speed setting when the minimum speed is set to zero. There may be significant variation in the minimum speed setting if the minimum speed is set higher than zero.

When adjustment of the controller board is complete, it is necessary to reattach the controller box front cover. Slide the front cover in place to cover the controller box and re-use the two (2) #6-32 screws to fasten the front cover in place to the control box.

Read these simplified instructions before operating.

Important: **Verify that the Dual Voltage Switch is set to the correct AC line input voltage, “115” or “230.”**

- Install the correct Plug-In Horsepower Resistor® according to input voltage and motor horsepower.
- Install proper size Armature Fuse.
- Install Auxiliary Heatsink on controls used with motors rated above $\frac{3}{4}$ HP on 120V AC, and 1- $\frac{1}{2}$ HP on 240V AC.
- Recheck connections: AC line to L1 and L2; armature to A+ and A-; and field (Shunt motor only) to F+ and F-. Connect ground via ground screw. (Note: if motor runs in improper direction, reverse armature leads.)
- Nominal trimpot settings are as follows (expressed in % of full clockwise rotation):

MIN (minimum speed)	15%
MAX (maximum speed)	60%
IR (IR compensation)	15%
CL (current limit/torque)	65%
ACCEL (acceleration start)	20%
DECEL (deceleration)	20%

For more detailed information refer to instructions in the previous pages or refer to the KBMM™ Installation and Operation Manual, provided by the D.C. board manufacturer